Predicting Vehicle Insurance Premiums with Machine Learning

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Introduction

This project explores factors influencing **vehicle insurance premiums** using machine learning, particularly **random forest** and **multiple linear regression** models.

With transportation being essential yet risky, vehicle insurance is crucial for financial protection. Both individuals and businesses seek fair premium rates, balancing affordability and coverage. By identifying key predictors, the analysis aims to support **insurance companies**, **policymakers**, and **consumers** in **ensuring fair pricing** and **informed decision-making**.

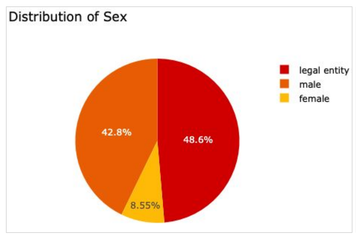
Dataset

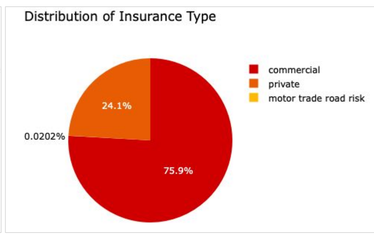
- Available on **Mendeley Data** online repository
- Obtained from the **Ethiopian Insurance Corporation**
- Data from **2014 to 2018**
- Data Size: **16 Features and 508,499 rows**

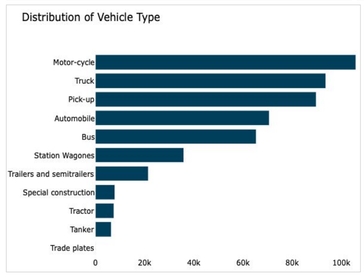
Key features:

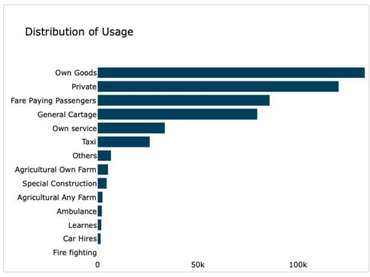
- Demographics: **Gender**
- Policy Information: **Insurance Period, Effective Year, Insurance Type, Insured Value**
- Vehicle Information: **Manufacturer, Vehicle Usage, Engine Capacity, Production Year**

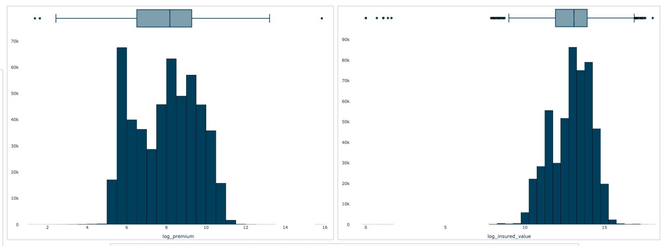
Exploratory Data Analysis

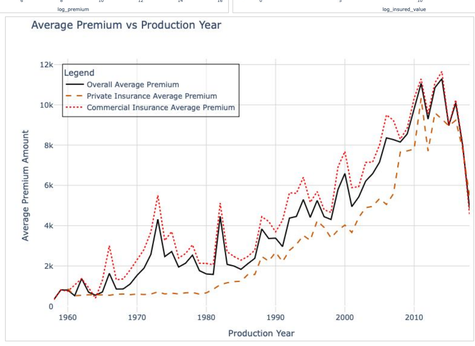






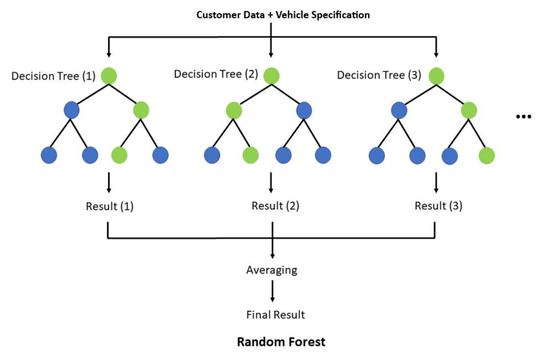






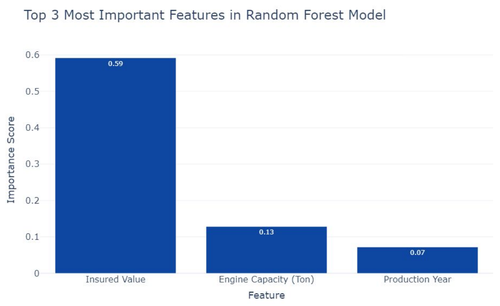
Random Forest

- Why Random Forest
  - A **flexible machine learning algorithm** suitable for diverse business scenarios
  - Ability to **capture non-linear relationships**
  - Provides feature importance score to **identify key pricing factors**

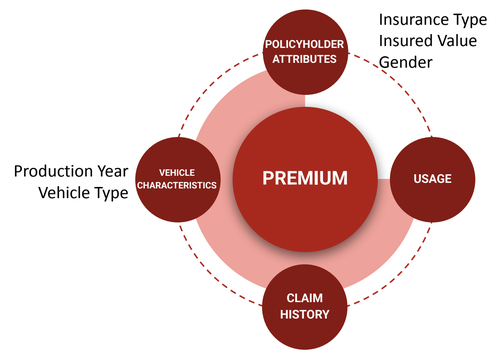


Random Forest

- Random Forest in the Insurance Industry
  - Risk Assessment & Pricing
  - Fraud Detection
  - Customer Retention & Churn Prediction
- Random Forest in this project
  - **Insurance premium prediction**
  - **Key influencing factors analysis**



Multiple Linear Regression



1. **Data Cleaning & Train-Test Split**
   - Missing values
   - Encode categorical variable
   - 80% / 20% split
2. **Model Fitting**
   - Additive Model
   - Influential Outliers
   - Interaction Terms
   - Polynomial Terms
3. **MLR Assumptions Testing**
   - Linearity
   - Normality of Residuals
   - Homoscedasticity
   - Multicollinearity
   - Independence of Errors
4. **Hyperparameter Tuning**
   - Lasso
   - Ridge
   - Elastic Net

- **MLR Application:** Predicted vehicle insurance premiums, identified key factors.
- **Issues:** Underfitting, multicollinearity in additive model.
- **Improvements:** Removed outliers, applied robust standard errors.
- **Overfitting:** Interaction and polynomial terms worsened it.
- **Regularization:** Lasso, Ridge, Elastic Net tested; Ridge performed best.
- **Final Model:** **Ridge regression** improved accuracy, reduced overfitting.
- **Key Insight:** Assumption testing, feature selection, regularization crucial for refining models.

Results & Discussions

Model Comparison

Model	Train R^2	Test R^2	Train Test Difference
Random Forest (Optimized)	78.30%	74.20%	5.5%
MLR (Ridge)	46.16%	42.24%	8.5%

- **Challenges**
  - Data quality issues
  - Model assumption violations
- **Limitations**
  - Bias-variance tradeoff
  - Computational challenges
  - Limited flexibility
  - Data sensitivity
- **Takeaways**
  - For Consumers
    - Choose a lower insured value that matches your risk profile
    - Choose a smaller vehicle
  - For Insurance Companies
    - Adopt machine learning models to improve premium pricing accuracy

Contact Information



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